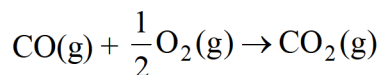
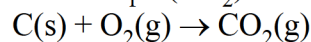


HINTS & SOLUTION

Q.1 (A)



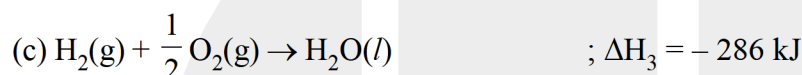
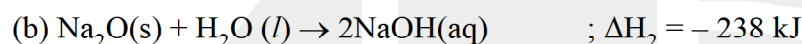
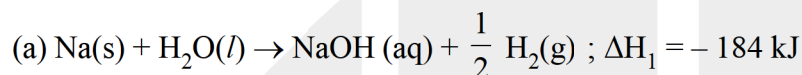
$$-280 = \Delta_f H(\text{CO}_2) + 120 \Rightarrow \Delta_f H(\text{CO}_2) = -400 \text{ kJ/mole}$$



$$\Delta_f H(\text{CO}_2) = 720 + 500 - 2 \times 710 = -200 \text{ kJ/mole}$$

Q.2 (C)

Given thermochemical equations are :



The required thermochemical equation is :



$$\text{From } 2 \times \text{(a)} + \text{(c)} - \text{(b)}, \quad \Delta H = 2 \times (-184) + (-286) - (-238) = -416 \text{ kJ}$$

Q.3 (A)

Q.4 (B)

 $\Delta H_{\text{formation}}$ of element in nature state is zero. ΔH_f of O_3 is endothermic ΔH_f of ZnO is exothermic & ΔH_f of $\text{H}^+_{\text{(aq)}}$ is zero.

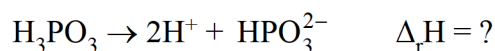
Q.5 (A)

$$\Delta H = [6\text{BE}_{\text{C-H}} + 2\text{BE}_{\text{C-C}} + \text{BE}_{\text{C}\equiv\text{C}}] - [3\text{BE}_{\text{C-C}} + \text{BE}_{\text{C}=\text{C}} + 6\text{BE}_{\text{C-H}}]$$

$$\Delta H = 835 - [348 + 610]$$

$$\Delta H = -123 \text{ kJ}$$

Q.6 (B)



$$-106.68 = \Delta_{\text{ion}} H - 55.84 \times 2$$

$$\Delta_{\text{ion}} H = 5 \text{ kJ/mol}$$

Q.7 (D)

$$\begin{aligned}\Delta H &= \Delta H_1 \times \frac{4}{5} + \Delta H_2 \times \frac{1}{5} \\ &= 100 \times \frac{4}{5} + 200 \times \frac{1}{5} \\ &= 80 + 40 = 120 \text{ cal}\end{aligned}$$

Q.8 (B)

$$\begin{aligned}nc\Delta T &= \Delta H_{\text{neutralisation}} \times \text{gm eq. neutralised} \\ 200 \times c \times t_1 &= \Delta H \times 20 \\ 600 \times c \times t_2 &= \Delta H \times 30 \\ \frac{t_1}{t_2} &= \frac{2}{3} \times \frac{3}{7} = 2\end{aligned}$$

Q.9 (A)



$$\frac{|\Delta_f H_{\text{Ag}^+(\text{aq})}|}{|\Delta_f H_{\text{Br}^-(\text{aq})}|} = \frac{5}{6}$$

$$|\Delta_f H_{\text{Ag}^+(\text{aq})}| = \frac{5}{6} [\Delta_f H_{\text{Br}^-(\text{aq})}]$$

$$\text{Let } \Delta_f H_{\text{Br}^-(\text{aq})} = +x \text{ kJ}$$

$$\Delta_f H_{\text{Ag}^+(\text{aq})} = -\frac{5}{6}x \text{ kJ}$$

$$\Delta H_{\text{reaction}} = \Delta H_{\text{f AgBr(s)}} - [\Delta H_{\text{f Ag}^+} + \Delta H_{\text{f Br}^-}]$$

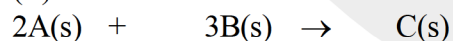
$$-100 \text{ kJ} = -110 \text{ kJ} + \frac{5}{6}x - x$$

$$+10 \text{ kJ} = -\frac{x}{6}$$

$$x = -60 \text{ kJ}$$

$$\Delta H_{\text{f Ag}^+(\text{aq})} = +\frac{5}{6} \times (60) \text{ kJ} = +50 \text{ kJ}$$

Q.10 (B)



$$\begin{aligned}\Delta H &= 2\Delta H_{\text{combustion of A(s)}} + 3\Delta H_{\text{combustion of B(s)}} - \Delta H_{\text{combustion of C(s)}} \\ &= 2(-200) + 3(-300) - (-400) \\ &= -900 \text{ kJ}\end{aligned}$$

Maximum heat will be released when A and B are taken in mole ratio as ratio of their stoichiometric coefficient (i.e. 2 : 3)

Let moles of A be 2x

moles of B be 3x

$$2x \times 200 + 3x \times 300 = 2600 \text{ g}$$

$$x = 2$$

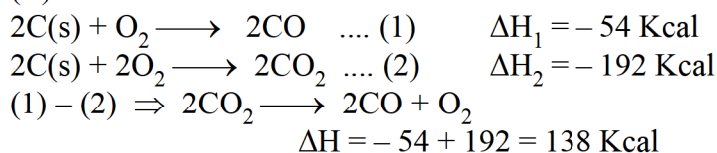
Moles of A taken = 4 ; Moles of B taken = 6

$$\text{Heat released} = 900 \times 2 = -1800 \text{ kJ}$$

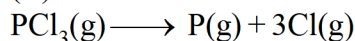
(900 kJ is released when 2 mole of A react with 3 mole of B)



Q.11 (C)



Q.12 (C)



$$3\text{BE}_{\text{P-Cl}} = 195 \text{ kcal/mol}$$

$$\text{BE}_{\text{P-Cl}} = 65 \text{ kcal/mole}$$

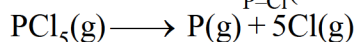
equatorial

in PCl_5

3P-Cl bonds are equatorial

$$\text{BE}_{\text{P-Cl}}(\text{eq}) = 65 \times 1.1 = 71.5 \text{ kcal}$$

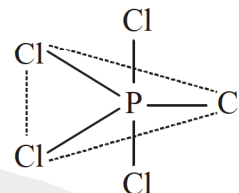
$$\text{BE}_{\text{P-Cl}}(\text{axial}) = 65 \times 0.9 = 58.5 \text{ kcal}$$



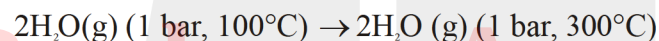
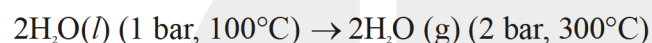
$$3\text{BE}_{\text{P-Cl}} + 3\text{BE}_{\text{P-Cl}}(\text{axial}) = 3 \times 71.5 + 2 \times 58.5 = 331.5 \text{ kcal}$$

(equatorial)

$$\Delta H = 331.5 \text{ kcal.}$$



Q.13 (C)



II

$$\text{In I: } \Delta H_1 = 2\Delta H_{\text{vap.}} = 2 \times 10.8 \text{ kcal} = 21.6 \text{ kcal}$$

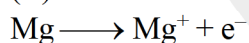
$$\text{In II: } \Delta H_2 = 2C_{p,m} \Delta T = 2 \times 4R \times 200 = 3.2 \text{ kcal}$$

(For $\text{H}_2\text{O(g)}$ triatomic (non linear): $C_{V,m} = 3R$, $C_{p,m} = 4R$)

$$\text{In III: } \Delta H_3 = nC_{p,m} \Delta T = 0$$

$$\text{Total change in enthalpy} = 21.6 + 3.2 = 24.8 \text{ kcal}$$

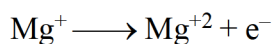
Q.14 (C)



To ionise one mole of Mg completely 750 kJ is spent and energy left = 550 kJ.

1450 kJ ionises 1 mole of Mg^{+1}

550 kJ ionises 0.38 mole of Mg^{+1}



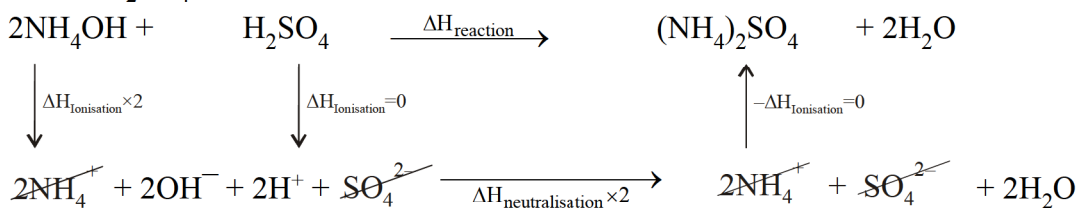
1

$$1 - 0.38 \quad 0.38$$

% of $\text{Mg}^+ = 62\%$ and that of $\text{Mg}^{+2} = 38\%$

Q.15 (C)

$$\text{mmole H}_2\text{SO}_4 = 0.1 \times 500 = 50 \text{ mmole}$$



as NH_4OH is 30% ionised to only 70% of it is to be converted into ions.

$$\Delta H_{\text{reaction}} = \Delta H_{\text{ionisation of WB}} \times 2 \times \frac{70}{100} + \Delta H_{\text{neutralisation}} \times 2$$

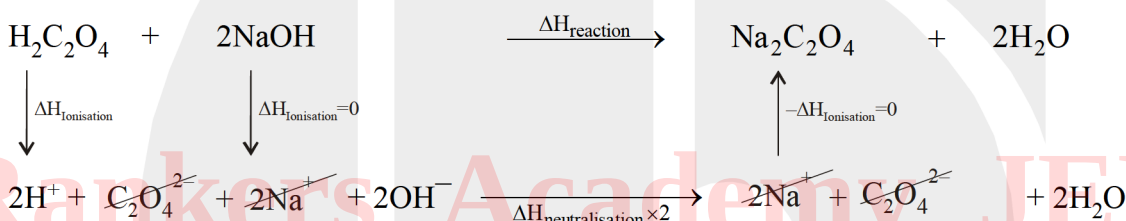
$$= 10 \times 2 \times \frac{70}{100} - 57.5 \times 2$$

$$\Delta H_{\text{reaction}} = -101 \text{ kJ for unit extent of reaction}$$

Thus heat released for neutralization of 50 mmole of H_2SO_4

$$\Delta H = -101 \times \frac{50}{1000} = -5.05 \text{ kJ}$$

Q.16 (C)

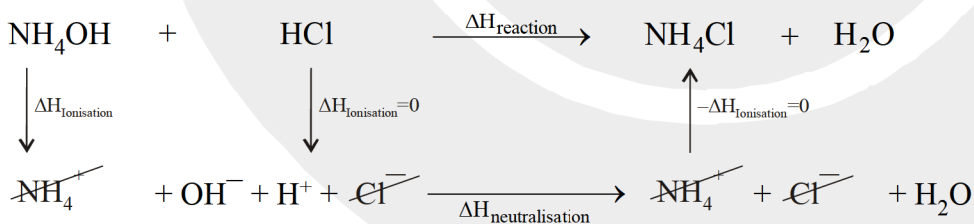


$$\Delta H_{\text{reaction}} = \Delta H_{\text{ionisation H}_2\text{C}_2\text{O}_4} + \Delta H_{\text{neutralisation}} \times 2$$

$$\Delta H_{\text{reaction}} = +20 \text{ kJ} - 57.5 \times 2$$

$$\Delta H_{\text{reaction}} = -95 \text{ kJ unit extent of reaction it involve two mole of NaOH & 1 mole of H}_2\text{C}_2\text{O}_4 \text{ for 1 mole of H}_2\text{C}_2\text{O}_4 \Delta H_{\text{neutralisation}} = -95 \text{ kJ}$$

Q.17 (C)



$$\Delta H_{\text{reaction}} = \Delta H_{\text{ionisation WB}} + \Delta H_{\text{neutralisation}}$$

$$\Delta H_{\text{reaction}} = (+10 - 57.5) \text{ kJ for unit extent of reaction} = -47.5 \text{ kJ}$$

$$\text{mole of NH}_4\text{OH} = 0.2$$

$$\text{mole of HCl} = 0.2 \quad \text{No limiting reagent}$$

$$\text{Released heat} = 0.2 \times 47.5 \text{ kJ} = 9.5 \text{ kJ}$$

$$Q_p = msdT$$

$$9.5 \times 1000 \text{ J} = 3000 \text{ ml} \times 0.95 \text{ gm/ml} \times \frac{1}{3} \text{ J/gm} \cdot \text{K} \times dT$$

$$dT = 10$$

$$T_2 = 310 \text{ K}$$



Q.18 (D)

ΔH° = enthalpy change at 1 bar pressure and at fixed temperature.

It depends upon temperature

By Kirchoff's equation

$$\left[\frac{\Delta H_2^\circ - \Delta H_1^\circ}{T_2 - T_1} = \Delta(C_p)_{\text{reaction}} \right]$$

$\Rightarrow$ For an ideal gas, enthalpy of a fixed amount depends only on temperature.

$$H = U + PV = U + nRT$$

$$\Downarrow \quad \Downarrow$$

$$\therefore H = f(T) \quad f(T) \quad f(T)$$

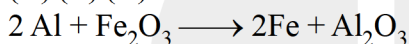
Q.19 (A) (C) (D)

$$(A) \Delta H_{\text{atomization, H}_2\text{O(s)}}^\circ = \Delta H_{\text{sublimation, H}_2\text{O(s)}}^\circ + 2\Delta H_{\text{BE, O-H}}^\circ$$

$$(C) \Delta H_{\text{atomization, H}_2\text{O(s)}}^\circ = -\Delta H_{\text{formation, H}_2\text{O(s)}}^\circ + \Delta H_{\text{BE, H-H}}^\circ + \frac{\Delta H_{\text{BE, O=O}}^\circ}{2}$$

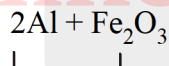
$$(D) \Delta H_{\text{atomization, H}_2\text{O(s)}}^\circ = \Delta H_{\text{fusion, H}_2\text{O(s)}}^\circ + \Delta H_{\text{atomization, H}_2\text{O(l)}}^\circ$$

Q.20 (A) (B) (D)



$$\Delta H_{\text{reaction}}^\circ = [(2 \times 0) - 390] - [(2 \times 0) - 176] = -214 \text{ kJ / mole}$$

For maximum value of heat,



moles 2 : 1

mass 54 : 160

Vol. 20ml : 50 ml

$$\text{Maximum calorific value of the fuel} = \frac{214 \times 1000}{214} = 1000 \text{ J/gm}$$

$$\text{Maximum calorific value of the fuel} = \frac{214 \times 1000}{70} = \frac{21.4}{7} \text{ kJ / mol}$$

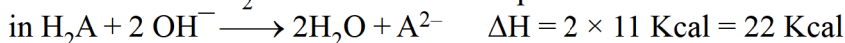
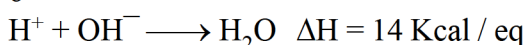
$$\text{For 2.14 kg mixture maximum heat} = 1000 \times 2.14 \times 1000 = 2140 \text{ kJ}$$

$$2500 \text{ kJ} > 2140 \text{ kJ}$$

$$\text{For 140 ml of mixture maximum heat} = \frac{21.4}{7} \times 140 = 428 \text{ kJ}$$

$$280 \text{ kJ} < 428 \text{ kJ}$$

Q.21 6



the heat theoretically released should be

$$2 \times 14 \text{ Kcal} = 28 \text{ Kcal}$$

thus $[28 - 22] = 6 \text{ Kcal}$ used for ionisation of weak acid H_2A



Q.22 0500

$$\text{Heat evolved per mol of 'B' atoms} = \frac{5}{2} \times 200 = 500 \text{ kJ}$$

Q.23 272 kJ/mol

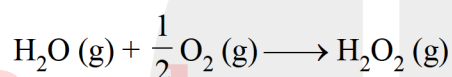
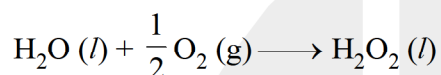
$$-847 = \left[31 + \frac{149}{2} + \frac{155 \times 5}{2} \right] - [5 \epsilon_{\text{I-F ax.}}]$$

$$\epsilon_{\text{I-F ax.}} = 268$$

$$-470 = \left[31 + \frac{149}{2} + 155 \times \frac{3}{2} \right] - [2 \times 268 + \epsilon_{\text{I-F eq.}}]$$

$$\epsilon_{\text{I-F eq.}} = 272 \text{ J}$$

Q.24 142

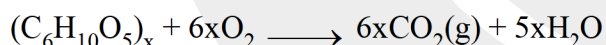


$$\Delta H = \left\{ \frac{1}{2} \times 498 - x \right\}$$

$$44 + 249 - x - 53 = -188 + 286$$

$$\Rightarrow x = 142$$

Q.25 1500



$$-4 \times 162x = -6x \times 94 - 5x \times 65 - 4 - \Delta_f H$$

$$\Delta_f H = -243x \text{ Kcal/mole}$$

$$\text{Per gm} = \frac{24300x}{162x} = 1500 \text{ cal/gm}$$

Q.26 $-1560 \text{ kJ mol}^{-1}$

Q.27 -75 kJ/mole

Q.28 $C = 12.68 \text{ J/mole-K}$, $W = 0$, $\Delta U = -31.7 \text{ J}$

Q.29 -350 kJ mol^{-1}

Q.30 $E_{\text{B-H-B}} = 455 \text{ kJ/mole}$

